

TABLE OF CONTENTS

OUR TEAM	1-2
OUTREACH	3-5
FINANCIAL PLAN	6
OUR ROBOT + GAME STRATEGY	7
ENGINEERING PROCESS	8
CHASSIS + FIRST INTERACTION	9
PIVOT AND LIFT	10
CLAW AND GUIDE SYSTEM	11
POWER TAKE-OFF SYSTEM	12
HANGING SYSTEM	13
PROGRAMMING	14-15

WHO ARE WE?

We are **Eastern Foxes 19098**, a robotics team from **Ploiești, Romania**. We were founded in **2020** from the desire to develop the STEAM field in our local community. Through our programs, we aim to **promote innovation, teamwork, and professionalism**.

COMPETITION EXPERIENCE

We began our journey in the Ultimate Goal Season (2020–2021) with participation in the regional stage. In Freight Frenzy (2021–2022), we were **Innovate Award Winners** at regionals and qualified for nationals. During PowerPlay (2022–2023), we were **Think Award Winners**, advancing again to nationals. In CenterStage (2023–2024), we won **Design Award 2nd Place** at regionals and **Motivate Award 2nd Place** at nationals. Also we qualified for the Maryland Tech Invitational, where we ranked **10th worldwide**. Now, in Into the Deep (2024–2025), we're proud to be **Control Award Winners** and **Finalist Alliance Captains** at both regional and national stages.

OUR TEAM



- Graphic Design
- Social Media
- Mechanical
- Programming
- Peer Mentor
- Financial
- Event Organization
- 3D Design
- Drive Team
- Mentor

OBJECTIVES

This season, we set clear goals across all areas of our team. From a **technical** standpoint, we focused on building a **fast robot** with **consistent autonomous** performance. On the **non-technical** side, we aimed to **double our sponsors** and improve **financial sustainability**, while also increasing the quality and **impact of our events** and promotional efforts. In terms of **team management**, we prioritized **preventing burnout**, optimizing our work time, **attracting recruits**, and fostering continuous internal growth. All efforts are made with the bigger picture in mind: preparing for the FIRST Championship, the highest level of competition.

FIRST AMBASSADORS

We have proven our abilities as FIRST ambassadors by promoting STEAM education. We launched the **FoxAid** series, with projects such as **Vision Beyond Limits** and **PawsAbility**, and provided **international mentoring** for **FTC Italy** and **FIRST Global winning team Madagascar**.

Our mission goes beyond competition as we strive to **inspire youth worldwide** to embrace technology as a tool for positive change.

As ambassadors, we share our passion aiming to continue **expanding our outreach**, encouraging young minds to dream bigger.

RECRUITMENT PROCESS

We prioritize **recruiting 9th and 10th grade students** to ensure team continuity and long-term development. We organize a two-day recruitment process to introduce participants to the FIRST Tech Challenge.

On the first day, we provide a **general presentation** of the team, roles, and the season's theme, followed by **two workshop options** in areas such as programming, mechanical engineering, graphic design, 3D CAD, web development, and event planning. Each workshop includes a **hands-on task**.

On the second day, candidates attend a **15-minute interview** where they can apply for departments like programming, mechanical engineering, public relations or graphic design, supporting their application with relevant portfolios.

The following week, we host **team-building sessions** that focus on both technical (e.g., construction, programming, 3D design) and non-technical skills (e.g., event planning, communication, management), helping new members integrate into the team and gain a deeper understanding of the competition.



ATTRACTING NEW MENTORS

Attracting mentors is essential for the continuous growth of our team. Many of our alumni choose to stay involved as active mentors, offering not only technical knowledge, but also strategic support. Each year, we look for new mentors who share our vision and are eager to contribute to developing new talent. This year, we are fortunate to have two new peer mentors, excited to support the team and help grow the STEAM community. The number of mentors vary, reflecting the team's evolution and the identification of new needs for further development.

TEAM STRUCTURE AND ORGANIZATION

Our team is divided into technical and non-technical departments, each playing a crucial role in our overall success.

Non-Technical Department

Social Media – Manages team accounts and creates engaging content

Events – Plans and organizes team events

Financial – Handles budgeting and sponsorship relations

Graphic Design – Manages the visual identity of the team, including social media graphics, merchandise, and promotional materials

Secretary Role – Ensures smooth internal organization by sending reminders, writing meeting minutes, organizing files, and managing communication platforms

Technical Department

Robot Programming – Develops and maintains the code that powers our robot

Web Development – Manages and updates the team website

3D Design – Creates digital models and technical drawings

Mechanics – Assembles, maintains, and improves the robot's hardware

INTERNAL COMMUNICATION

Internally, we stay connected via WhatsApp groups, while online meetings are held on Discord or Google Meet, depending on our needs. We also use two Facebook groups to store and share important materials and meeting records, ensuring no essential information is lost.

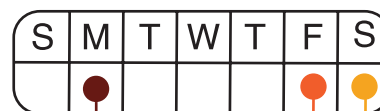
HOURS SPENT AT THE TEAM HUB

This chart represents the distribution of hours spent at the team hub.

It highlights the collective effort and energy invested in achieving the team's goals.

WEEKLY MEETING SCHEDULE

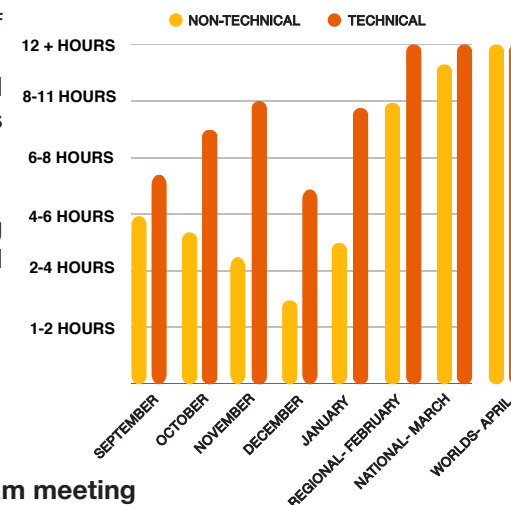
We follow a well-structured meeting schedule to maintain productivity and alignment across all departments.



Technical department

Non-technical department

Full team meeting



OUR VALUES THROUGH EVENTS

Fox Aid Series



INNOVATION

A series that constantly inspires us to make a difference. Among the projects we developed are Vision Beyond Limits, a device designed to help visually impaired people navigate, created in collaboration with a local association; and Paws Ability, a motor-support orthosis for animals, implemented with the help of a local veterinary clinic. We plan to continue expanding this series, as it shows us how to apply robotics to real-life challenges.



INCLUSION

"Robotics for the Future"

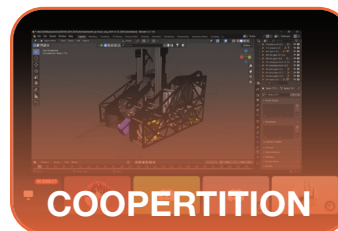
One of our dearest events, which proved that every activity truly matters. We organized interactive workshops for students from both the Middle School and the Theoretical High School "Filipești de Pădure", reaching over 13 classes and more than 400 children. The activities included LEGO and Arduino sessions, as well as presentations about the FTC competition. Students also had the opportunity to interact with our robot and even become a "driver for a day."



IMPACT

Eastern Community Lounge

An event dedicated to STEAM education that brought together 23 clubs and organizations from Ploiești, alongside 12 inspiring speakers who shared their experiences in the fields of science and technology. The event reached over 2,000 people from the local community.



COOPERTITION

"All for One and One for All"

Our international series of meetups with teams from different countries, reached over 30 teams. We learned how the FIRST Tech Challenge is experienced in other parts of the world, exchanged ideas, and built meaningful global connections.



TEAMWORK

Eastern Arena & WaveZ Arena

Two league-meet competitions organized by our team, where over 40 FTC teams had the opportunity to test and improve their robots in a competitive environment. For WaveZ Arena, we partnered with Team TehnoZ, strengthening valuable friendships within the FIRST community.



CITIZENSHIP

Eastern Green Week

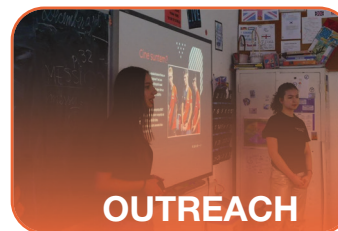
A week dedicated to ecology, aiming to show that robotics is not just about competition, it's also about social responsibility. We held educational activities and workshops for our high school students, as well as presentations alongside environmental experts.



DISCOVERY

Workshop Series

We host weekly Arduino workshops in collaboration with the "Nicolae Iorga" County Library, with over 20 regular participants. In addition, we organize specialized workshops for our 80+ volunteers and STEAM enthusiasts, covering topics such as entrepreneurship, leadership, 3D design, and more—having held over 40 sessions so far. Our partnership with LEGO Romania also enabled us to run 4 LEGO-based workshops in their locations in Ploiești and Bucharest.



OUTREACH

Eastern on the Road

This season, we have significantly expanded our activities through our school promotion caravan, reaching 25 schools in our county. Our goal is to double this number by the end of the school year, promoting our team and the FIRST community.

FUTURE PLAN

Expanding the caravan nationwide, reaching at least half of the schools in our county and nearby areas.

Diversifying the workshop themes to attract new STEAM enthusiasts and form future team members.

Developing projects from the **FoxAid series** by installing and testing new devices for orientation in specialized centers.

Optimizing the recruitment process by extending the selection period to 3-4 days, thus providing more time for integrating new members and achieving a more **efficient recruitment rate**.

Increasing the number of training sessions for team members, ensuring better preparation and continuous development.

Participating in international competitions such as the Maryland Tech Invitational, Chicago Robotics Invitational, and Bucharest Twin Cup.

TRAINING SESSIONS

We take part in **training programs** offered by **Leaders and Fedevo**, where we learn effective teamwork, explore artificial intelligence, and apply essential marketing concepts.

Each year, some of our members **join Erasmus+ VET mobilities**, gaining hands-on experience in electronics, Java programming and Arduino, which we later apply in our team projects.

We've attended **training sessions** with professors from the **University of Oil and Gas Ploiești**, who supported us in improving our programming skills, technical writing, business understanding, and promotional strategies.

Through the **"Nație prin Educație"**, the official partner of FIRST in Romania, **mentorship program**, we focused on **team management** and **event organization** with the guidance of **Bianca Olaru and Andreea Ghiță**, enhancing both our internal coordination and the quality of our events.

We also work closely with our **mentors** throughout the season, who guide us in solving technical challenges and developing both individual and team-wide skills.

TRAINING SESSIONS WITH THE HELP OF EXPERTS



MIHAELA TĂTARU – County Youth Foundation

- Supported us in implementing community-based projects like the FoxAid series
- Helped us obtain funding for projects, including Arduino and LEGO kits sponsored by our local community foundation



LIVIU DRĂGHICI – Author, Teacher & Entrepreneur

- Through his entrepreneurial experience, he guided us in self-analyzing our skills and better understanding our roles within the team



DANA VASILESCU – FEDEVO, Youth Development Association

- Helped us understand the concept of emotional intelligence, which proved useful during TV interviews and in managing our emotions before important events



Professor DOREL DUȘMĂNESCU – University of Oil and Gas Ploiești

- Assisted in developing the engineering process
- Helped ensure the coherence of technical concepts in our documentation
- Supported us in applying physics concepts to our projects



IOAN TOPÎRCEAN

- Guided our team in understanding the CNC manufacturing process
- Taught us about material strength and their optimal use in building components



MIHAI NICOLAE – “Observatorul Prahovean” – Local TV Station

- Promoted our events within the community
- Helped share our story online and create successful articles using specific journalism techniques to capture the reader's attention and effectively convey information

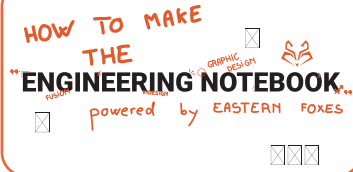
The Impact of Professionals on Our Team

Connecting with professionals in the fields of robotics and STEAM has helped us develop valuable skills in various domains, thus laying the foundation for our continuous learning and growth journey.



EMPOWERING TEAMS AS FIRST AMBASSADORS

The Engineering Notebook Guide



This guide has supported teams nationwide by outlining the key steps to building a successful engineering notebook, both technical and non-technical. It tackles common challenges like business plan structure and required diagrams.

Total: 530 views

The Programming Guide



The Programming Guide was initially created to support rookie teams participating in FTC Italy. Later, we decided to publish the guide to help newly formed teams around the world learn the basics of FTC programming. The guide was also accompanied by 2 online sessions for better understanding.

Total: 423 views

Reach: 35 participants in the training sessions



After creating the Programming Guide, our collaboration with FTC Italy expanded through regular online meetings with the 20 teams. We monitored the progress of their programming departments and offered guidance to help them continuously improve.

Mentoring the Madagascar Team – FIRST Global



We mentored the team representing Madagascar, who went on to win FIRST Global. Through weekly sessions, we supported them in developing their robot, specifically the take-up extension, and offered guidance on fundraising, notebook structure, and design principles.

“FTC Beyond the Field” podcast



Our podcast aims to help teams better understand the FIRST competition. The podcast's guests are FTC alumni, who share their experiences and offer advice for some of the biggest challenges. The podcast had a total of 6 episodes for each area of the competition: events, portfolio and notebook, programming, construction, interview and 3D design.

Total: 15000 views

TEAM AND EVENT PROMOTION

This year, we implemented a **well-structured promotion strategy** across multiple platforms, including **Instagram, Facebook, TikTok, and X**. With a **consistent posting schedule** organized in a **shared Google Calendar**, we doubled our reach compared to previous seasons. **Planning ahead** allowed us to **save time** and remain consistent, while every piece of content was carefully designed to reflect the values of FIRST and represent our team through **eye-catching graphics and meaningful messages**.

Beyond online presence, all our events were developed under the identity of our brand, **EASTERN**, with flagship initiatives like Eastern Green Week and Eastern Arena. This **branding approach** helped us build a strong, **recognizable image within our community**.

Our **promotional materials** were another key asset in our outreach strategy. With a wide variety of items including **stickers, business cards, flyers, notebooks, wristbands, and badges**, we attracted hundreds of people to our booths at various events. These materials not only **expanded our visibility** but also emphasized the importance of visual identity. At nationals, we saw this in action: a well-designed stand made people stop, engage, and remember us.

One of the most **innovative elements** in our promotion strategy was a **billboard** placed in the center of our city, which featured our team, recent achievements, and upcoming events a bold step that helped amplify our presence locally.

We also worked with several **local and national TV stations and news platforms**, which regularly featured our results and activities. These media appearances significantly increased our audience and event attendance, with many people learning about our story and mission through televised interviews and coverage. Additionally, we used our robot to bring the ball for the kick-off of the soccer match in our city. During the match, we also made a presentation, **reaching over 15,500 people** and helping them understand what the FIRST Tech Challenge competition is about.



15 TV APPEARANCES

83 ARTICLES

INSTAGRAM
1,785 followers
Total views 780k

FACEBOOK
450 followers
Total views 130k

YOUTUBE
160 subscribers
Total views 6400

TIK TOK
985 followers
Total views 2.3M

SUSTAINABILITY PLAN

Sponsors and Partners

We have 8 recurring sponsors who have been supporting us since the PowerPlay season, ensuring a stable financial foundation. Each season, we also collaborate with additional companies that support us depending on their availability.

Diversifying Funding Sources

We access STEAM grants and funding opportunities, such as the "Științescu" Fund, which provided us with a \$2000 grant. With this, we purchased LEGO and Arduino kits that we used to organize weekly workshops for children in our local community.

Sponsor Relationships

For us, our supporting partners are a key pillar in the team's development. That's why we focus on maintaining a strong and transparent relationship with them.

We send out an every two weeks newsletter to keep them informed about our progress, the events we participate in, and how we promote their brand both online and through visual materials such as t-shirts, roll-ups, and at our pit area.

To show our appreciation and strengthen our partnerships, we host the Eastern Business Lounge an biannually event dedicated to our sponsors and partners. It provides an ideal setting for networking, building long-term connections, and attracting new supporters who share our vision of innovation and technological progress.

Sponsorship Timeline

12+ 10+ 17+ 8

New sponsors March to September

New sponsors September to March

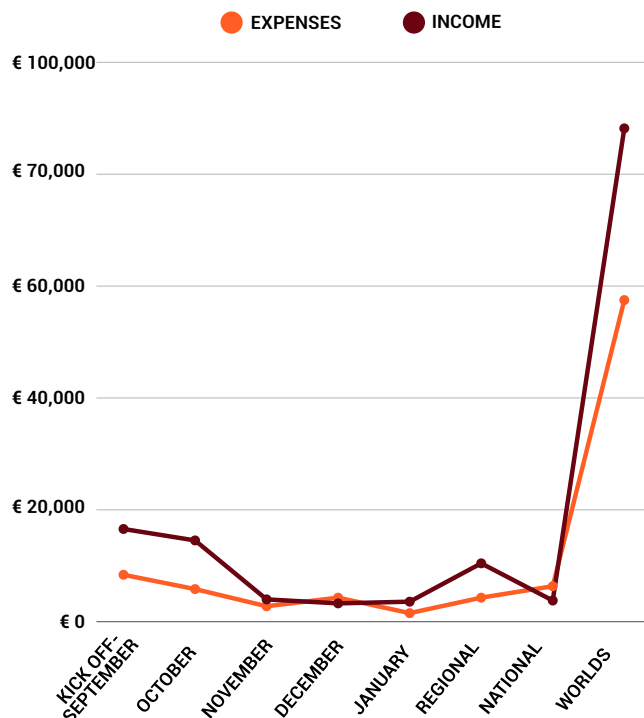
New sponsors for Worlds

Recurring sponsors

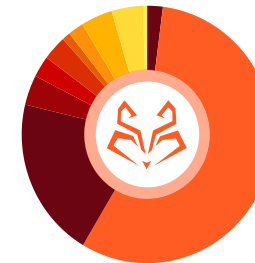
Total of 47 sponsors this season

FINANCIAL PLAN

REPRESENTATION OF EXPENSES AND INCOME IN EUROS



BUDGET ALLOCATION OVERVIEW



The annual budget was distributed across several key categories: the largest portion, **59.27%**, was allocated to **Transportation & Accommodation**, followed by **18.92%** for **Robot Parts**.

Equipment and Hub Maintenance each received **3.78%**, while **Promotional Materials & Merch** accounted for **3.53%**. **Field & Basic Parts** took **2.90%**, and the **Pit Area** was allocated **2.52%**. **Monthly Subscriptions**, which include 3 Adobe Creative Cloud accounts, Canva, Envato, and ChatGPT-4, made up **1.89%**. **Consumables** also took **1.89%**, with **Events** at **1.01%** and **Other expenses** at **0.50%**.

FUTURE INVESTMENTS

We aim to further develop the STEAM field, which is why we plan to expand our workshop series by acquiring new kits.

We invest in organizing high-quality events that attract a larger audience thanks to the excellent conditions we offer (event catering, snack bar, relaxation area).

To increase our visibility, we focus on both online and community-based promotion. We invest in online advertising for our events, distribute promotional materials (flyers, stickers, business cards), and make ourselves known through a billboard placed in the city center. This strategy has helped us double the number of event participants and significantly grow our community on social media.

MAIN SPONSORS



MAIN PARTNERS





OUR ROBOT "VELOX"

Our robot, named "Velox", features an **ultra-lightweight chassis**, a **fast lift** powered by three motors, an **innovative PTO**, and a **quick and reliable claw**. Its on-field performance is driven by both **innovative weight reduction** solutions and various automations, such as **collection failsafes**, **guide sensors**, and a **camera** for autonomous collection from the Submersible.

GAME STRATEGY

During the 2-minute **teleoperated phase**, our drive team controls the robot to complete objectives based on **real-time game developments**. We **adapt** our strategy dynamically in coordination with alliance partners. After aligning plans, we conduct a **joint practice session** to ensure smooth collaboration. We have developed **three core strategies**:

- Specimen Strategy** – If the allied robot is more efficient at basket scoring, our robot **feeds** samples to the human player, who converts them into specimens. Then, we **collect and place** specimens on the **high chamber**. With the remaining time, we place the yellow sample in the high basket and ascend to level 3.
- High Basket Strategy** – Our robot collects and scores **samples** in the **high basket**. When it is **full**, we move to the low basket. The final 7 seconds are dedicated to a level 3 ascend.
- Mixed Strategy** – If the ally lacks key capabilities, we **begin** with basket scoring while they **feed** samples. Mid-match, **roles switch** to maximize efficiency. We finish with a level 3 ascend.

ROBOT IN NUMBERS



3.9 Optimal Cycle Time



90% Custom Parts



306 Solo Highscore



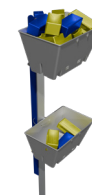
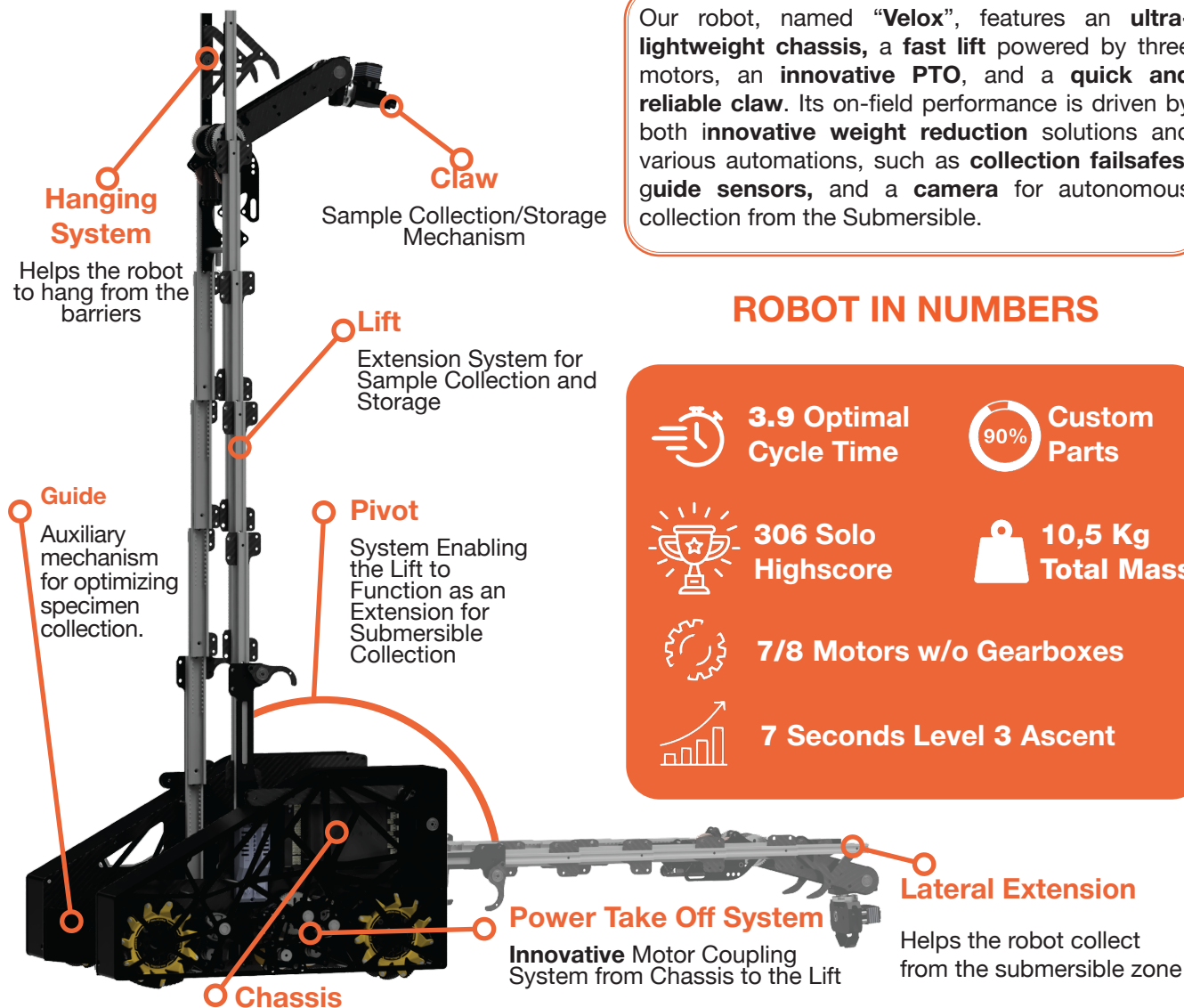
10,5 Kg Total Mass



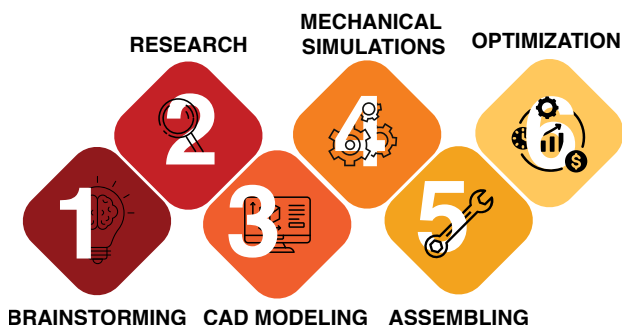
7/8 Motors w/o Gearboxes



7 Seconds Level 3 Ascent

 $x 21 = 168 \text{ pt.}$  $x 13 + 7 = 174 \text{ pt.}$  $x 14 = 140 \text{ pt.}$ FIRST
TECH
CHALLENGEINTO THE
DEEP2024
SEASON #9
2025ORGANIZATOR
NATIE
PRIN EDUCAȚIEPARTENER FONDATOR
BRD
GROUPE SOCIETE GENERALE

ENGINEERING PROCESS



Brainstorming : The team comes together to think up the **most efficient solutions** for the season's challenges

Research : Analysing **past seasons** for what systems worked best and how we could implement them

CAD Design : Putting all of our ideas together in CAD to better **visualise and problem solve**

Mechanical Simulations : Testing the **structural integrity** of our systems before bringing them to life

Prototype Assembly : **Manufacturing** the robot itself

Optimizations : **Testing** and **improving** the subsystems of the robot for the **best field performance**

THE EVOLUTION AFTER EACH COMPETITION

First League Meet - Hansei WaveZ:

The robot performed as expected. Both the chassis and mechanisms were very fast.

Encountered Issues: The robot was **unstable** when placing in the basket due to the lift.

Solutions:

We printed **two extra mounts for the slides** as support for stability.

Second League Meet - Eastern WaveZ:

The collection time was improved by implementing **automation** with the help of a **camera**. Scoring failures were **reduced** by reinforcing the linkage system, which significantly enhanced the lift's stability.

Encountered Issues:

Loss of points due to the **lack** of a hanging system.

Solutions:

Manufacturing hooks and **redesigning** the climbing system.

Third League Meet - Starbloom:

Implementing the climbing system **increased our average scores**. With experience, we began to move more efficiently and successfully navigate difficult situations. However, **due to tolerance** variations, the autonomous collection was not always precise.

Encountered Issues:

Lack of **climbing on the second bar** and **autonomous inconsistency**. Climbing on the first bar was relatively slow.

Solutions:

Changing the **autonomous routine** to a new one, where sample placement to the human player is done using the chassis.

Redesigning the climbing system.

Fourth League Meet - Eastern Arena:

The redesign of the climbing system allowed us to score the **15 ascend points** in just **2 seconds**.

Encountered Issues:

Difficulties in time management during the **endgame period**.

Solutions:

More hours of **practice**.

League Tournament:

We refined both the autonomous and collection systems to improve overall performance. The robot was **tested extensively** to ensure stable and reliable operation.

Encountered Issues:

The **lack of a level 3 ascent** costed us valuable points in the endgame.

The **lack of a sample autonomous routine** made us **incompatible** with some of our alliance partners in the qualification matches.

Solutions:

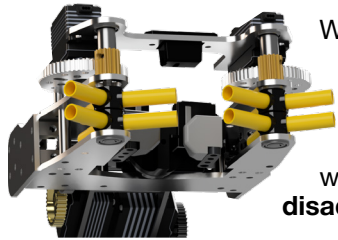
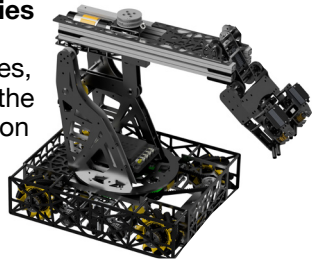
We added a **level 3** climbing auxiliary system.

Implemented an **autonomous routine for basket scoring**.



FIRST ITERATION OF THE ROBOT

Observing the similarities between the **Into The Deep** and **Freight Frenzy** challenges, we adapted the systems of the best robots from that season to test them in the current competition. Our research began by **analyzing** the dominance of the **active intake** during that season, aiming to implement a similar system. Alongside this, we **experimented** with a **turret system** to explore different scoring methods.

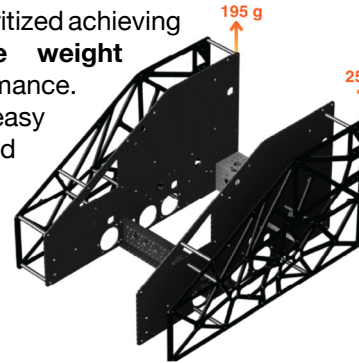


With this robot, we participated in the **first competition of this season**, the WATT's UP Challenge 2 demo, where we identified the main **disadvantages of this design**:

The **robot's center of gravity** was **too high**, making it unstable when extended.

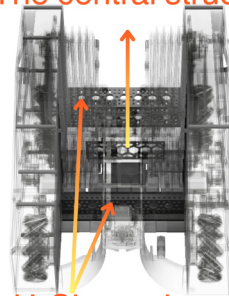
The **intake** was **too heavy and bulky** to be operated solely with servomotors.

The chassis design prioritized achieving the **lowest possible weight** to optimize performance. However, to facilitate easy access to cables and the PTO, we chose **carbon fiber** for **inner plates**, and opting for **aluminum side plates** with **weight-reduction holes**.



To **enhance lateral stability**, we connected motors from opposite sides, effectively stiffening the robot's structure.

The central structure



U-Channel

$$F = \frac{T}{r} = \frac{1.83}{0.052} = 35.30N$$

$$a = \frac{F}{m} = \frac{35.30}{19.5} = 3.36m/s^2$$

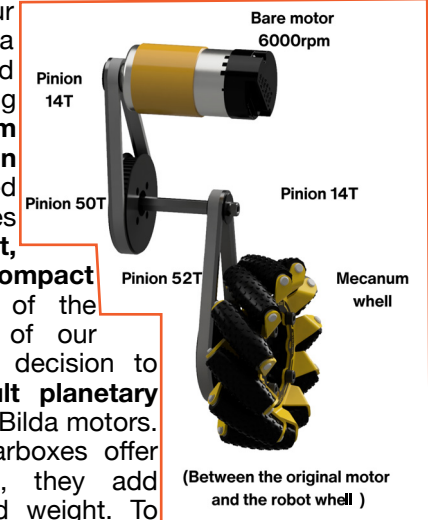
$$m = 10.5$$

$$r = 0.052m$$

$$T = 18.7 \frac{N \cdot m}{cm} = 1.83Nm$$

The **most significant improvement** made to maintain the robot's low weight was **removing the gearboxes from 7 of the 8 motors**. This **reduced the total weight substantially**—by 580 grams—as we eliminated gearboxes and implemented custom gear ratios, significantly optimizing efficiency and weight.

For this season, our team engineered a highly optimized robot by developing a **fully custom lift and drivetrain system** designed around the principles of **minimal weight, high speed, and compact form factor**. One of the **core innovations** of our approach was the decision to **remove the default planetary gearboxes** from GoBilda motors. Although these gearboxes offer torque advantages, they add significant bulk and weight. To address this, we designed our **own custom gear reduction system** using GT2 belts and pulleys, which allowed us to tailor **specific gear ratios** for different subsystems while significantly reducing the size and weight of each assembly. This modular, **lightweight** solution enabled us to create a compact drivetrain and allowed more freedom for internal component placement.



The formula to calculate the transmission ratio is:

$$\text{Transmission Ratio} = \frac{\text{Engine RPM} \times \text{Small Pinion Teeth}}{\text{Large Pinion Teeth}}$$

Given the parameters:

Engine RPM = 6000rpm, Small Pinion Teeth = 14, Large Pinion Teeth = 50

Substituting the values into the formula:

$$\frac{6000 \times 14}{50} = \frac{84000}{50} = 1680 \text{ rpm}$$

So, the output RPM after the first stage of gearing is 1680 rpm. Now, for the second calculation:

$$\frac{1680 \times 14}{52} = \frac{23520}{52} = 452 \text{ final rpm}$$

CHASSIS

We strived for our chassis to meet **three criteria for optimal performance**:



Stability



Speed



Reliability



FIRST
TECH
CHALLENGE

INTO THE
DEEP

2024
SEASON #9
2025

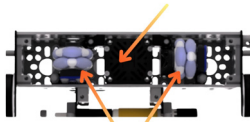
ORGANIZATOR
NATIE
PRIN EDUCAȚIE

PARTENER FONDATOR
BRD
GROUPE SOCIETE GENERALE

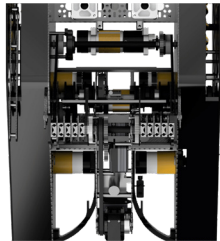


We integrated the **odometry system** into the chassis, assuring protection and a stable mounting position. Alongside odometry, we also implemented an external Pinpoint IMU, offloading some calculations from the Control

GoBilda Pinpoint



Odometry Deadwheels

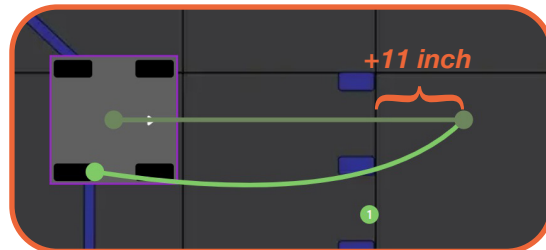


Sweeper

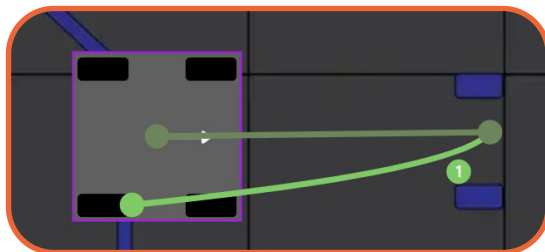
Hub and significantly reducing loop times.

Another chassis optimization was the **addition of a "sweeper"**, enabling quicker Sample collection by shortening autonomous trajectories. This mechanism

allows the robot to push Samples toward the human player, effectively reducing the robot's total travel distance during autonomous routines.



Path followed during the first autonomous routine



Path followed during the second autonomous routine

PIVOT

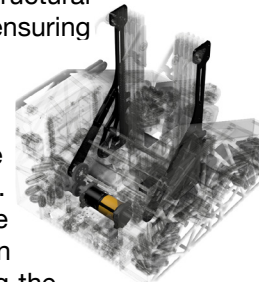
The pivot was designed to meet **essential criteria**:

Reliability

Narrow Design

Speed

The pivot's design merges structural simplicity with high reliability, ensuring both precise and stable operation, all while maintaining a slim profile that integrates seamlessly into the robot's overall architecture. Additionally, the pivot allows the lift to function as an extension mechanism, effectively removing the need for a separate transfer system. This dual functionality not only saves space but also reduces mechanical complexity and potential points of failure.



LIFT

The lift system consists of **10 Misumi SAR230 slides**, arranged in two groups of 5 rails on each side. We incorporated carbon fiber inserts to enhance rigidity. This setup significantly reduces vibrations and deformation under load. The lift, powered by **three motors**, can rapidly raise our 800g intake mechanism at a speed of 4.02 m/s. A **REV encoder** on the motor shaft ensures precise real-time positional feedback. Additionally, the lift doubles as an extension, mounted on a pivoting system.



Issue : Instability and excessive weight of the lift.

Solution : We lightened each slide by drilling holes into them to reduce the robot's overall weight and, at the same time, to lower its center of gravity.

$$r = \frac{48}{2} = 24 \text{ mm} = 0.024 \text{ m.}$$

Calculation of the Cable Linear Speed

The angular speed of the drum is:

$$\omega = \frac{1600 \times 2\pi}{60} \text{ rad/s}$$

$$\omega = \frac{1600 \times 6.2832}{60} = 167.55 \text{ rad/s.}$$

The linear speed of the cable is given by the relation:

$$v = \omega \cdot r.$$

Substituting the numerical values:

$$v = 167.55 \times 0.024 = 4.02 \text{ m/s.}$$

Thus, the linear speed of the system before considering losses is **4.02 m/s**

FIRST ITERATION

We used a carbon fiber mechanism powered by a motor. The motor's power is transmitted to the pivoting mechanism through gears.

Issue: During testing, this version exhibited **instability** due to **excessive flex** in the carbon fiber plates.

SECOND ITERATION

Solution: In this version, we integrated **aluminum alongside carbon fiber components**. This combination ensures enhanced rigidity, reducing vibrations and deformation under load, thereby **maintaining structural integrity under stress**.

CLAW

The claw system was designed to meet **essential** criteria:

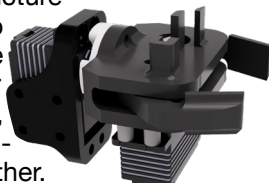
 Lightweight

 Degrees of Freedom

 Simple Design

FIRST ITERATION

The claw has a 3D printed structure that supports both the servo motor and the claw itself. One claw finger is driven by another directly mounted on the servo, both fingers featuring semi-gears that **engage** each other.



The entire collection system is mounted on a **mobile four-bar mechanism**, pivoting 270 degrees between the position for **collecting** specimens from the guide and the position for collecting samples from the submersible.

Issue : Slow collection

SECOND ITERATION (FINAL)

Solution : We've implemented a solution to **assist drivers in faster collection from the submersible**, optimizing both the precision and efficiency of the process. A **small camera on the claw** provides better visibility of the target objects and facilitates their detection. By integrating an **AI-based detection software** which we developed, we were able to **analyze** the position of the elements in **real-time** and **automatically align the claw** for more **accurate collection**. This reduces driver error, accelerates the collection time, and improves **operator feedback**. Additionally, the system allows dynamic adjustments based on **environmental conditions**, ensuring stable and efficient operation even in challenging situations.

GUIDING SYSTEM

Issue : Collecting **specimens from the perimeter** is **too slow**.

Solution : Implementing a mechanism that allows us to collect specimens more quickly.

The guide was designed to help us quickly and efficiently collect specimens from the **human player**. Following our research, we found perimeter collection to be **too slow** for our needs, so we developed a mechanism to improve our cycle time.

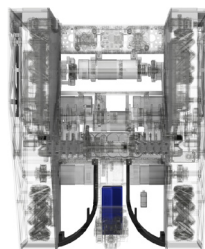
In designing this mechanism, the guide had two main objectives:

 Fast intake

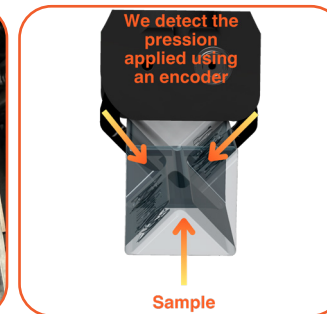
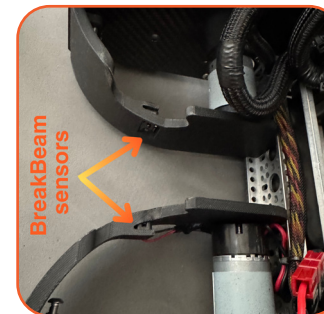
 Reliability

 Automation

The guide was designed to allow the specimen to enter with a **small margin of error**. It is rounded inward, ensuring that the robot pushes and aligns the specimen upon contact. Inside the guide, on both sides are **break-beam sensors**, which detect when the game element is placed in the desired position and initiate the collection sequence.



We utilize **two unique modes** to ensure reliable collection during both the Tele-Op and Autonomous periods. With the help of **two high-precision break-beam sensors**, the robot detects when the specimen is oriented correctly into the guide, initiating the collection sequence. Additionally, we have implemented another **failsafe mechanism**: an **absolute encoder** that accurately recognizes when a sample has been successfully collected.



POWER TAKE OFF SYSTEM

A Power Take-Off (PTO) system is a mechanism used to transfer additional power from a chassy motor to another subsystem of the robot. The system is designed to **engage only when needed**.

The PTO system was developed to **integrate** the **climbing mechanism** into the robot's lift. During the design process, **three essential** criteria were prioritized:

 Efficiency

 Reliability

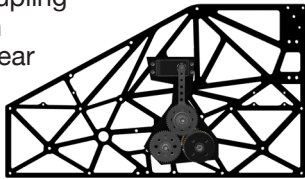
 Lightweight

WHY DO WE NEED THIS SYSTEM?

To **lift** the robot using the slides, we must balance **optimal** torque with speed to **score** as many samples as possible during a match. To maintain the lift extension speed at the desired level, it was necessary to **innovate** a system that provides **additional torque** for lifting the chassis. After analyzing the success of **various** hanging mechanisms from previous seasons, we chose to implement the **PTO system** due to its simplicity and **ease of integration** into our current robot.

FIRST ITERATION

The first solution for the PTO system involved implementing a forced coupling between two gears. When activated, a shaft with a gear mounted on it coupled to another one mounted on the chassis motors and simultaneously engaged another gear connected to the lift motors' shaft. This mechanism was operated via a linkage driven by a servo motor, which ensured the coupling of the free gear to the rest of the system.



This solution had two main **disadvantages**: Efficiency losses due to the use of three gears in sequence;

The stress placed on the servo motor, which maintained the system in the coupled position.

$$P = T \times \omega$$

$$\omega = \frac{2 \times \pi \times N}{60}$$

$$\omega = \frac{2 \times 3.1416 \times 435}{60} \approx 45,53 \text{ rad/s}$$

$$P = 300 \times 45,53 \approx 13.659 \text{ W} \approx 13,66 \text{ kW}$$

$$P = 400 \times 45,53 \approx 18.212 \text{ W} \approx 18,21 \text{ kW}$$

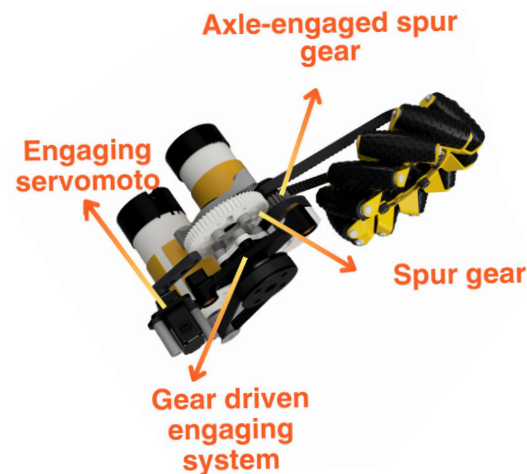
SECOND ITERATION(FINAL)

The second version of the PTO system has the advantage of a simple, efficient design and mechanical reliability. The current system addresses the drawbacks of the initial prototype, eliminating servo coupling stress and removing the risk of gear skipping.

We extended the shaft responsible for the lift spools, and on the newly created segment, a gear was mounted that can rotate freely around it. This gear is connected to one of the chassis drive motors.

A fixed element is firmly attached to the shaft, unable to rotate but with the freedom to slide along its length. The element is kept under tension by two torsion springs, which push it in the direction of the gear. A servo motor acts as the locking mechanism, preventing the fixed element from sliding.

The fixed element and the gear on the shaft have contact points in the form of bolt heads that intersect when the two components come into contact. This configuration allows for efficient power transmission from the chassis motors to the lift shaft, compensating for the torque required to lift it.



$$T_{pto} = 2 \cdot T_{sasiu} = 2 \cdot 13.56 = 27.12 \text{ kg/cm}$$

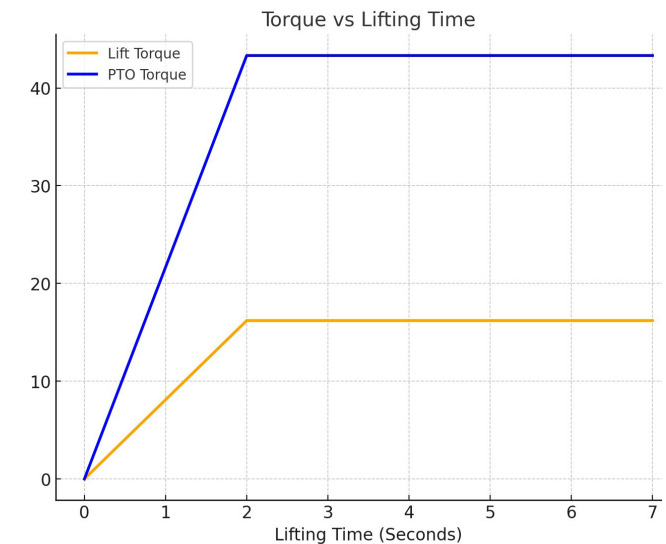
$$T_{total} = T_{pto} + T_{lift} = 27.12 + 16.2 = 43.32 \text{ kg/cm}$$

$$Addition = \frac{T_{total} - T_{lift}}{T_{lift}} \cdot 100$$

$$Addition = \frac{43.32 - 16.2}{16.2} \cdot 100$$

$$Addition = \frac{27.12}{16.2} \cdot 100$$

$$Addition = 164.4\%$$





HANGING SYSTEM

In the design of the climbing system, the following criteria were **prioritized**:



FIRST ITERATION

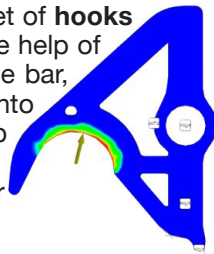
The **hooking system** consists of two-stage arms, a bare arm and one with hooks at the end. An 8mm shaft is mounted from one side to the other of the chassis plates and functions as a **spool** for the hook arm. It is driven by a servo motor, run with a 4:3 gear ratio. The **two-stage arms** are kept under tension by elastics, which assist their **instant extension** to the desired level. This design worked to **ensure** our climb on the first bar, but it limited our ability to perform a level 3 climb.

Issue : Slow hang

SECOND ITERATION

Solution: We decided to use **motors** to lift the robot faster, utilizing the **power of two chassis** motors, with the help of a PTO system. A set of **hooks** are mounted on the lift, and with the help of the pivot, they are positioned on the bar, lifting the robot onto it. Once lifted onto the bar, a set of **mobile hooks** grab onto the bar.

Issue: The mechanism was under stress during transfer.



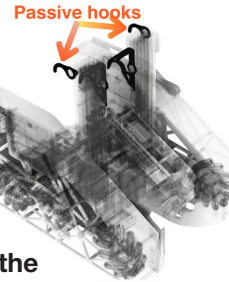
THIRD ITERATION (FINAL)

Solution :

We added an **auxiliary system** consisting of two bars operated in the same way as the first system, **pushing against the first hanging rack** to move the robot from under it. Along with this system, **we replaced the fixed hooks** on the lift with a set of **passive hooks** to ease the transfer.

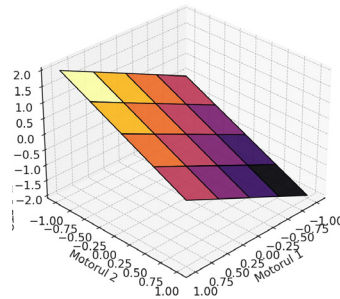


Transfer assist system



Passive hooks

Since climbing using the lift puts **considerable stress** on the cable responsible for operating it, we **implemented** a system that forces the **return springs** not to exceed a **preset length** to **prevent permanent deformation**.



Mechanical endstop

Total Force Required for Lifting

The total force required to lift the robot is given by:

$$F_{\text{total}} = m_{\text{total}} \cdot g$$

where: - $m_{\text{total}} = 10.8 \text{ kg} + 2 \cdot 0.2 \text{ kg} = 11.2 \text{ kg}$ is the total weight of the robot (including the hooks), - $g = 9.81 \text{ m/s}^2$ is the gravitational acceleration.

Thus, the total force is:

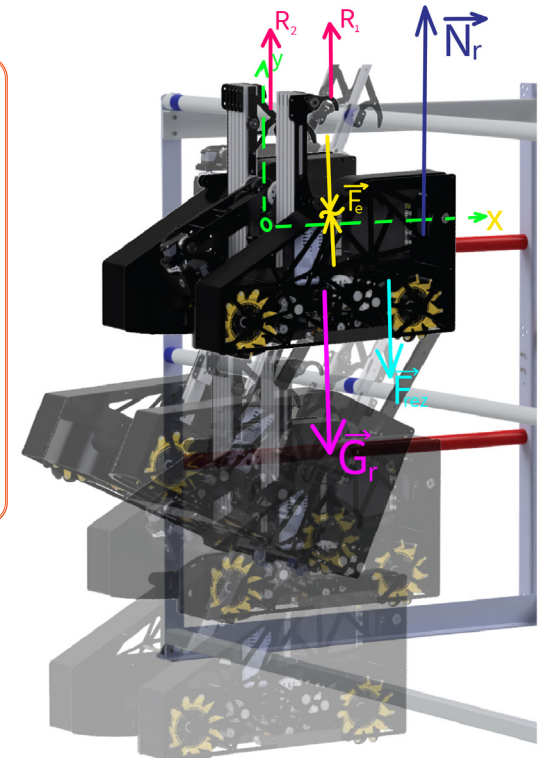
$$F_{\text{total}} = 11.2 \text{ kg} \cdot 9.81 \text{ m/s}^2 = 109.87 \text{ N}$$

Force Distribution Across Slides

Assuming the force is evenly distributed across the 10 slides, the force per is:

$$F_{\text{per slide}} = \frac{F_{\text{total}}}{n_{\text{slides}}} = \frac{109.87 \text{ N}}{10} = 10.99 \text{ N}$$

The forces present during hanging



Transition Ascent Level 1 - Level 3



Finite State Machine and Update in TeleOp Loop

First, in our code, we apply the kinematic **equation** for the robot's drive train, which is essential in developing the program that handles individual subsystems of the robot. Each subsystem is modular, and every subsystem works through a **dedicated class**.

For a structured program, the main class of **TeleOp** initializes specific controllers for each subsystem of the robot. Thus, each component is initialized separately, maintaining a clear architecture and ease of extension. The **initialization** of these subsystems starts with the INIT function. This method allows for the separate management of each subsystem, facilitating the management of each individual component.

To efficiently manage the behaviors of each subsystem, these classes use a **Finite State Machine (FSM)**. This approach allows each subsystem to change its behavior depending on current conditions, ensuring more robust and predictable control of the robot.

Clear Structure & Modularity

Each subsystem has its own independent behavior

Easy Debugging

Each subsystem can be individually tested and improved.

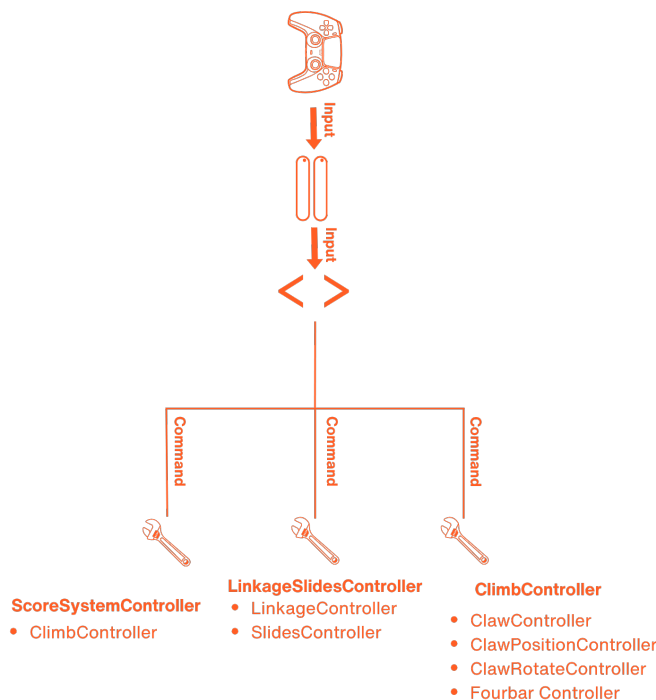
Rapid Response Efficiency

State changes for subsystems are managed rapidly after initialization

Scalability

New subsystems can be added without affecting the rest of the code.

With these programming concepts as the foundation, we managed to complete the drive with optimizations using various sensors and general automations.



Sensors and General Automations

We use a Break Beam sensor to automatically collect specimens from the observation zone. It functions by sending a boolean-type signal, influencing whether the sensor's receiver is hit or not by the emitter's range.

The odometry connected to the Pinpoint system (which contains a gyroscope) allows localization during autonomy.

Control Systems using PID

A common practice in robotics and mechanics is when we have to make components move against external forces (such as gravity), implementing a PID (Proportional – Integral – Derivative) control system.

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt}$$

We use a total of six PID controllers: one for the robot's pivot, one for each of the three motors on the linear axes, and the system that handles periods of autonomy.

Autonomy: Strategy&Control

The League Meet-structure determined that we need to maximize the AUTO score in the league ranking. Therefore, our focus was on consistent performance, as that's how we would be ranked, and we tested it in competitions.

In fig. 1, we illustrate the path of the robot as it scores 5 samples. These are the first competitive events of the season, and we collected sample values for the robot's movement in the field. However, performance is dependent on consistency, so we used elements of observation while using this data to discuss with the team the configuration. During the TeleOp period, we focused more on controlling the robot with the Human Player input, testing our Auto scoring and subsequent TeleOp phase.

In fig. 2 and 3, we show the trajectory of the robot as it scores a total of 8 samples.

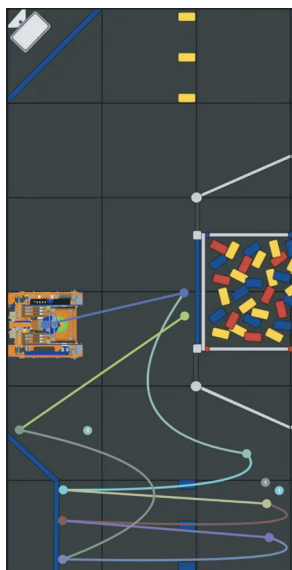


Fig 1

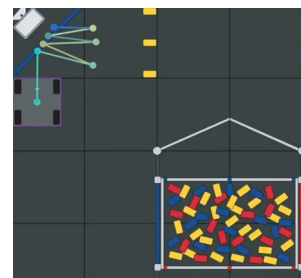


Fig 2

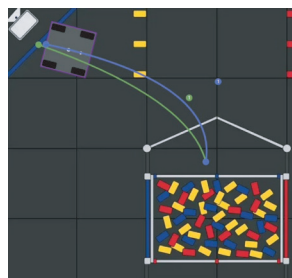


Fig 3

For movement during the autonomous period, I used the **Pedro Pathing** control system, an advanced vector follower developed by the FTC 10158 team. Unlike Roadrunner, it uses **Bézier curve generation** to produce smoother, faster, and more reliable trajectories.

These curves are mathematically defined by a sequence of **control points**, and the resulting trajectory is an interpolation between these points. To **generate** a Bézier curve of order n , the following formula is used:

$$B(t) = \sum_{i=0}^n P_i \cdot \binom{n}{i} \cdot (1-t)^{n-i} \cdot t^i, \quad t \in [0, 1]$$

Where:

$P_0, P_1, \dots, P_i$ are the control points, t is a parameter that varies between 0 and 1, determining the position on the curve, and $\binom{n}{i}$ is the binomial coefficient, which defines the contribution of each control point to the final curve.

Additionally, the 4 PID controllers we implemented using Pedro provide dynamic adjustment in case the robot bumps or gets stuck during autonomy.

The 4 implemented PID controllers are used for:

- **Translational PID**: Corrects the robot's lateral direction on the trajectory and ensures the chassis follows a straight path.
- **Heading PID**: Ensures the robot maintains its rotational alignment and corrects deviations from the desired rotation.
- **Drive PID**: Manages acceleration and braking along a path, ensuring smooth movement and minimizing overshooting of the desired coordinates.
- **Secondary Drive PID**: Acts at the end of a trajectory for smoother braking.

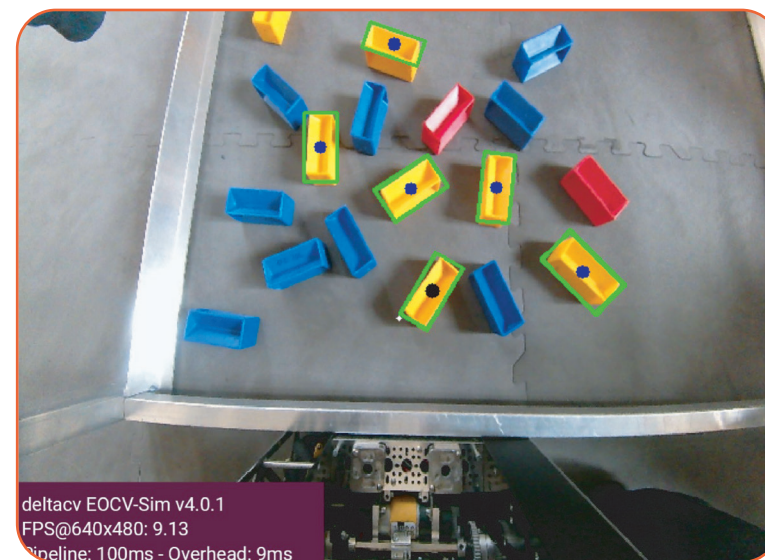
Using these PID controllers (or vectors), **Pedro Pathing** fulfills its function as a **reactive vector** follower. Pedro allows the robot to follow both a straight line and react to any external force that **disrupts the trajectory** by recalculating at each step. Since this system corrects itself in real-time, we save the time that other motion planning libraries would use to adjust the position at the end of the trajectory.

Sample detection during autonomy.

Our detection code is an **OpenCV pipeline** designed to **detect** and analyze yellow objects in the images captured by the robot's camera. It **converts** the image from RGB format to HSV for more reliable color detection, applies a segmentation filter to isolate regions of interest, and uses **morphological operations** to enhance the clarity of contours.

After the initial processing, the pipeline applies a **contour detection algorithm** and **filters** the results based on predefined criteria, such as the **area** and **aspect ratio** of the object. Then, the center of each valid contour is identified, and the closest point to the image center is selected as the **main reference**. For this point, the object's orientation angle is calculated, which is **essential** for the robot's **alignment** and **navigation**.

This pipeline is **optimized** for real-time execution and can be used for **autonomous adjustment** of the robot's direction in relation to a sample or visual marker. By **integrating** it into the control system, the robot can make **fast and precise decisions** during the competition, ensuring enhanced performance in scenarios requiring object **detection** and **tracking**.



This approach allows us to maintain **reliable** and **efficient** autonomy throughout the competitions.